

Report of Calibration  
PSC ALSu\_1ST\_ARTICLE\_2-CH-SLOW\_HSSS/N 0001  
2026-09-25 10:06:31

Calibration Current standard: BNL PSC ATE S/N 001  
Calibration Volt standard: HP 3458A-002 S/N 2823A 23647  
Calibration Resistance standard: Fluke 742A-1 S/N 1063008  
End Header

lab{2}Chan1  
Burden resistor = 33.3333

Measuring initial gains and offsets

| Itest      | dacSP     | dcct1      | dcct2      | dacRB     | err      |
|------------|-----------|------------|------------|-----------|----------|
| -1.005167  | 1.008580  | -1.015460  | -1.015422  | 1.002036  | 0.003933 |
| -27.002106 | 27.004295 | -26.818411 | -26.826691 | 27.060005 | 0.003633 |

|                           | dacSP     | dcct1     | dcct2     | dacRB    |
|---------------------------|-----------|-----------|-----------|----------|
| Initial measured offsets: | -0.003460 | -0.017794 | -0.017434 | 0.000000 |
| Initial measured gains:   | 0.999953  | 0.992538  | 0.992858  | 1.002395 |
| Gain corrections:         | 0.999953  | 1.007518  | 1.007193  | 1.000000 |

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

| DAC SP    | DAC RB    |
|-----------|-----------|
| 1.000000  | 0.997075  |
| 27.000000 | 27.058010 |

Measured offset: -0.005269  
Measured gain: 1.002344  
Gain correction: 0.997662

Writing gain and offset constants for dacRB to PSC

Verification

| Itest      | dacSP     | dcct1      | dcct2      | dacRB     | err      |
|------------|-----------|------------|------------|-----------|----------|
| -1.004898  | 1.005148  | -1.005044  | -1.005036  | 1.004789  | 0.033568 |
| -27.001982 | 27.002096 | -27.001894 | -27.001860 | 27.001375 | 0.030929 |

|                         | dacSP     | dcct1     | dcct2     | dacRB     |
|-------------------------|-----------|-----------|-----------|-----------|
| Final measured offsets: | -0.000256 | -0.000155 | -0.000148 | -0.000346 |
| Final measured gains:   | 0.999995  | 0.999991  | 0.999990  | 0.999986  |

|                               | dacSP     | dcct1     | dcct2     | dacRB     |
|-------------------------------|-----------|-----------|-----------|-----------|
| Final measured offsets mean:  | -0.000058 | -0.000154 | -0.000133 | -0.000121 |
| Final measured offsets stdev: | 0.000276  | 0.000114  | 0.000118  | 0.000262  |
| Final measured gains mean:    | 1.000005  | 0.999996  | 0.999996  | 0.999995  |
| Final measured gains stdev:   | 0.000012  | 0.000010  | 0.000010  | 0.000009  |

Saving channel 1 calibration constants to qspi

lab{2}Chan2  
Burden resistor = 33.3333

Measuring initial gains and offsets

| Itest      | dacSP     | dcct1      | dcct2      | dacRB     | err       |
|------------|-----------|------------|------------|-----------|-----------|
| -1.004690  | 1.008134  | -1.014933  | -1.014198  | 1.008812  | -0.029973 |
| -27.001575 | 27.011204 | -26.825111 | -26.817787 | 27.101786 | 0.047914  |

|                           | dacSP     | dcct1     | dcct2     | dacRB    |
|---------------------------|-----------|-----------|-----------|----------|
| Initial measured offsets: | -0.003205 | -0.017459 | -0.016978 | 0.000000 |
| Initial measured gains:   | 1.000238  | 0.992818  | 0.992565  | 1.003457 |
| Gain corrections:         | 1.000238  | 1.007234  | 1.007491  | 1.000000 |

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

| DAC SP    | DAC RB    |
|-----------|-----------|
| 1.000000  | 1.004333  |
| 27.000000 | 27.100452 |

Measured offset: 0.000637  
Measured gain: 1.003697  
Gain correction: 0.996317

Writing gain and offset constants for dacRB to PSC

Verification

| Itest      | dacSP     | dcct1      | dcct2      | dacRB     | err       |
|------------|-----------|------------|------------|-----------|-----------|
| -1.004617  | 1.004684  | -1.004709  | -1.004737  | 1.005255  | -0.017254 |
| -27.001843 | 27.001753 | -27.001751 | -27.001736 | 27.002136 | 0.043472  |

|                         | dacSP     | dcct1     | dcct2     | dacRB    |
|-------------------------|-----------|-----------|-----------|----------|
| Final measured offsets: | -0.000073 | -0.000099 | -0.000129 | 0.000578 |
| Final measured gains:   | 0.999994  | 0.999993  | 0.999991  | 0.999993 |

|                               | dacSP     | dcct1     | dcct2     | dacRB    |
|-------------------------------|-----------|-----------|-----------|----------|
| Final measured offsets mean:  | -0.000205 | -0.000123 | -0.000185 | 0.000240 |
| Final measured offsets stdev: | 0.000132  | 0.000134  | 0.000132  | 0.000312 |
| Final measured gains mean:    | 0.999988  | 0.999991  | 0.999990  | 1.000009 |
| Final measured gains stdev:   | 0.000012  | 0.000009  | 0.000009  | 0.000017 |

Saving channel 2 calibration constants to qspi

Test data reviewed by \_\_\_\_\_ Date\_\_\_\_\_

